

Volatility-Estimation Lag and Calibration Memory: A Sequential Prediction-Interval Feasibility Study

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Abstract

Sequential prediction intervals can respond slowly to a scale change for two reasons: the current scale forecast may lag, and earlier scale errors remain in the calibration scores after that forecast recovers. This report separates those mechanisms in a finite-window model and examines one controlled Gaussian scale-jump pilot. The archived pilot contains 200 independent paths. For a 250-score window, the causal EWMA-normalized rolling interval attained 95.24% coverage overall but 83.42% over the first 25 forecasts after the upward scale step; the raw rolling interval attained 94.28% and 53.72%, respectively. A separate deterministic contamination calculation shows how inflated historical scores can persist after the current scale becomes exact. These are development results from one synthetic design. The work does not establish a new method, a general conditional-coverage guarantee, or a novelty claim.

1 Introduction

This study concerns one-step intervals for a scalar next response Y_t under time-varying conditional scale. The question is whether scale normalization improves the immediate response to a volatility change and how long historical scale-estimation errors can affect a finite calibration window. The distinction matters because an interval can have near-nominal coverage averaged over a long evaluation period while missing many responses immediately after a shift.

Adaptive conformal inference (ACI) already studies online coverage under distribution shift and uses GARCH-normalized scores in a realized-volatility example [1]. Sequential predictive conformal inference (SPCI) predicts future residual quantiles using temporal dependence and gives an assumption-dependent asymptotic conditional-coverage analysis [2]. Strongly adaptive online conformal prediction and conformal PID control provide further adaptive calibration approaches [3, 4]. The present pilot implements none of those comparators. Its purpose is to isolate a candidate mechanism and specify the evidence needed for a later comparison, rather than to claim priority or superiority.

2 Sequential interval protocol

Let $\mathcal{F}_{t-1}$ contain the observations available before Y_t is revealed. At origin t , a deployable method uses only $\mathcal{F}_{t-1}$ to issue a location forecast $\hat{\mu}_t$, a positive scale forecast $\hat{\sigma}_t$, and an interval. After Y_t

is observed, it stores the *vintage* score

$$R_t = \frac{|Y_t - \hat{\mu}_t|}{\hat{\sigma}_t}.$$

The forecasts in an old score remain those issued at its original origin; they are not recomputed using a later model fit. In the pilot the mean is known to be zero. For a rolling window of W previous scores, let

$$k = \lceil (W + 1)(1 - \alpha) \rceil, \quad q_t = R_{(k); t-W:t-1}, \quad C_t = [\hat{\mu}_t - \hat{\sigma}_t q_t, \hat{\mu}_t + \hat{\sigma}_t q_t].$$

Here $R_{(k); t-W:t-1}$ is the k th order statistic of the saved scores. If $k > W$, the specified threshold is infinite. The current response never enters its own calibration window. Forecast cutoffs, unavailable scores, and failed fits must remain visible in a later full evaluation; the corrected rank alone gives no general coverage guarantee under temporal dependence.

3 Scale error and calibration memory

Suppose $Y_t = \mu_t + \sigma_t Z_t$ with exact location forecasts. If every forecast scale in a window, including the current one, equals the true scale times the same constant $c > 0$, then $R_i = |Z_i|/c$. The score quantile is $q_t = q_t^{\text{oracle}}/c$, and $\hat{\sigma}_t q_t = \sigma_t q_t^{\text{oracle}}$. Thus a stable multiplicative scale bias cancels from this rolling interval. Scale forecast error by itself does not determine interval error; the *change* in scale error across score vintages is relevant.

Write $\eta_i = \log(\hat{\sigma}_i/\sigma_i)$ and let G be the CDF of $|Z_i|$. Assume the innovations are identically distributed and continuous, each Z_i is independent of $\mathcal{F}_{i-1}$, and each η_i is fixed before Y_i is revealed. Define $H_i(x) = G(e^{\eta_i}x)$, its window average $\bar{H}_t(x)$, and the empirical score CDF $\hat{H}_t(x)$. When the W scores are distinct and $k \leq W$, the conditional coverage $c_t = H_t(q_t)$ obeys the pathwise decomposition

$$|c_t - (1 - \alpha)| \leq \underbrace{\sup_x |\hat{H}_t(x) - \bar{H}_t(x)|}_{A_t} + \underbrace{\sup_x |\bar{H}_t(x) - H_t(x)|}_{B_t} + \left| \frac{k}{W} - (1 - \alpha) \right|.$$

If G has density g and $L = \sup_{u \geq 0} u g(u) < \infty$, then $B_t \leq LW^{-1} \sum_{i=t-W}^{t-1} |\eta_i - \eta_t|$. This separates empirical fluctuation, relative scale-error drift, and rank rounding. It is a diagnostic decomposition, not an automatic finite-sample conditional-coverage theorem; the empirical term requires its own analysis.

For an exact, narrower calculation, assume the current scale has recovered. Among W independent historical scores, $W - m$ have distribution $U = |Z|$ and m have distribution cU . Membership in the two groups and c are fixed before the scores are drawn. Let q be their k th order statistic and let U_0 be an independent new draw. Conditioning on $U_0 = u$ gives

$$\Pr(U_0 \leq q) = \int_0^\infty \Pr\{A_u + B_u \leq k - 1\} dG(u), \quad \begin{cases} A_u \sim \text{Bin}(W - m, G(u)), \\ B_u \sim \text{Bin}(m, G(u/c)), \end{cases} \quad (1)$$

with independent binomial counts for each fixed u . Indeed, $U_0 \leq q$ exactly when at most $k - 1$ historical scores lie below U_0 , apart from probability-zero ties. For $m = 0$ or $c = 1$, the clean repeated-sample coverage is $k/(W + 1)$. Coupling the same U draws shows that $c > 1$ cannot lower this repeated-sample coverage, while $c < 1$ cannot raise it. This identity does not transfer unchanged to adaptive EWMA or fitted GARCH scores, whose errors depend on past observations. Its proof and relation to prior work still require independent scientific review.

4 Exploratory experimental design

The archived pilot draws 200 independent paths of length 2,600 with $Z_t \sim N(0, 1)$, known mean zero, and true scale $\sigma_t = 1$ until zero-based origin 1200, 3 for origins 1200–1799, then 1 again. Evaluation covers origins 1000–2599 at nominal coverage $1 - \alpha = 0.95$. The causal EWMA forecast starts at variance $v_0 = 1$ and updates only after reveal:

$$\hat{\sigma}_t = \sqrt{v_t}, \quad v_{t+1} = 0.94v_t + 0.06Y_t^2.$$

The principal comparison uses raw rolling absolute errors, EWMA-normalized rolling vintage scores, and an oracle-scale rolling diagnostic, all with $W = 250$. A full Gaussian oracle uses the true current scale and the normal 0.975 quantile. All methods share the same innovations within each path; oracle rows have privileged information and are not deployable competitors.

Coverage is averaged first across forecast origins within each path and then across paths. Its Monte Carlo standard error (MCSE) is the sample standard deviation of the 200 path-level means divided by $\sqrt{200}$. Mean width is reported relative to the full Gaussian oracle width at the same origin. Analytic conditional coverage in this Gaussian simulation is also averaged over paths for the event-time figure. These uncertainty summaries condition on this single fixed data-generating design, not a population of volatility processes. The archive also contains $W = 50$ variants; their clean rank benchmark is $49/51 = 0.9608$, versus $239/251 = 0.9522$ for $W = 250$, so their coverage difference cannot be interpreted as pure adaptivity.

5 Results

Table 1 shows that average coverage can mask a severe immediate post-shift deficit. For raw rolling calibration, the first 25 upward-shift forecasts cover 53.72% of outcomes; EWMA normalization raises this to 83.42% in this design but still falls well below 95%. The oracle-scale diagnostic is close to its clean finite-rank level. Following the downward step, raw rolling coverage over the first 25 forecasts is 100%, with mean width 3.03 times the full-oracle width; the EWMA-normalized counterpart has 99.98% coverage and width ratio 2.38. Neither observation establishes a method ranking against strong adaptive baselines.

Table 1: Archived 200-path pilot at $W = 250$. Coverage and MCSE are percentage points; width ratio is the mean interval width divided by full Gaussian oracle width. Both oracles use the true current scale.

Method	Overall (%)	MCSE	Up 0–24 (%)	MCSE	Width ratio
Raw rolling	94.283	0.027	53.72	0.570	1.195
EWMA-normalized rolling	95.242	0.015	83.42	0.355	1.079
Oracle-scale rolling	95.225	0.017	95.40	0.282	1.017
Full Gaussian oracle	94.962	0.041	94.90	0.301	1.000

Figure 1 shows a second view of the same upward shift using conditional coverage, which averages over the future Gaussian outcome while retaining variation in past histories. The plotted values are descriptive 25-origin bin means from the archived trajectory. They do not supply bin-level uncertainty intervals or a new confirmatory test.

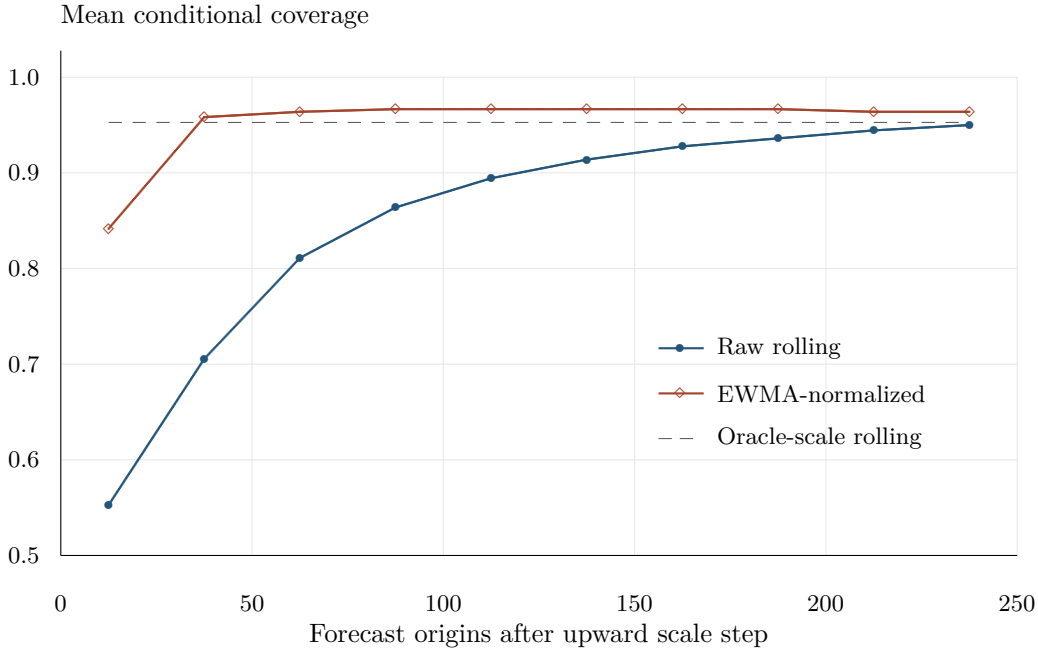


Figure 1: Mean conditional coverage in consecutive 25-origin bins after the upward scale step, averaged over the 200 archived paths. Points are placed at bin midpoints. Lines guide the eye; no bin-level MCSE is shown.

Table 2 evaluates the independent deterministic-score model in Eq. (1), not the EWMA pilot. With $W = 250$ and $k = 239$, 25 inflated historical scores at $c = 3$ raise repeated-sample coverage from the clean 95.219% to 98.955% despite an exact current scale. Replacing c by $1/3$ lowers it to 94.690%. The numerical quadrature and 40,000 independent Monte Carlo checks agree within the recorded Monte Carlo errors. These calculations illustrate a possible post-recovery memory effect; they do not measure how often a fitted scale model creates that pattern.

Table 2: Deterministic contamination model at $W = 250$, $k = 239$. Quadrature and Monte Carlo (MC) coverage are percentages; MCSE is in percentage points.

Contaminated m	Multiplier c	Quadrature (%)	MC (%)	MCSE
0	1	95.2191	95.2182	0.0067
25	3	98.9553	98.9564	0.0037
25	1/3	94.6903	94.6942	0.0074
50	3	99.9192	99.9187	0.0007

6 Limitations and next study

The pilot has one Gaussian scale-jump profile, one EWMA decay, a known mean, and development seeds only. It includes neither a fitted GARCH scale forecast nor ACI, SPCI, strongly adaptive online conformal prediction, or conformal PID. It uses no dated real series. Oracle methods know the true current scale at an unannounced change, so their performance is a diagnostic boundary, not an achievable forecast at that origin. Overall coverage, interval width, and event-time behavior

must be evaluated together; no result here establishes general conditional coverage or practical superiority.

The next study should first test Eq. (1) over a predeclared grid of contamination counts and multipliers, including both directions and clean controls. It should then add causally fitted scale forecasts and faithful adaptive comparators on common planned evaluation origins, retaining failed fits and infinite intervals. Hyperparameters, scenarios, primary contrasts, and confirmation seeds should be fixed before the confirmation runs. Mathematical assumptions and overlap with prior work need independent review before a new contribution is claimed.

7 Reproducibility

The repository includes the Python pilot, numerical identity checks, forecast-before-reveal tests, a locked dependency file, and the archived synthetic outputs. The numerical source for Table 1 is `reference/synthetic_baseline_v1/pilot_results/summary.csv`; Figure 1 aggregates `reference/synthetic_baseline_v1/pilot_results/trajectory.csv`; and Table 2 uses `reference/synthetic_baseline_v1/theory_results/coverage_identity.csv`. The six numerical archive files have recorded SHA-256 hashes in `reference/synthetic_baseline_v1/SHA256SUMS.txt`. This standalone source embeds the plotted values and requires no external data file to compile.

The archived 200-path pilot records Python 3.13.5, NumPy 2.3.5, SciPy 1.17.0, and seed 20260928. The current locked workspace uses Python 3.12.9. Its 56 Python tests passed; a fresh two-path development smoke run matched all 560 keyed per-replication metrics in the archived seed prefix at absolute and relative tolerance 10^{-12} . A full 200-path reproduction in the current workspace has not been completed. The two-path agreement checks implementation consistency for that prefix, not the research claims in this report.

References

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